VLSI Devices
Lecture 4

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L4

Remember the boundary condition.

- At the silicon-oxide interface,

$$\epsilon_{ox}|\mathbf{E}_{ox}| = \epsilon_{si}|\mathbf{E}_{si}|$$

Taur, Eq. (4.6)

  - Therefore, we have a non-zero "surface potential."

$$\phi_s = \phi(0) - \phi(\infty)$$

  - The overall potential difference is $V_g - V_{fb}$.
  - The oxide potential, V_{ox}
  - It is important to note that

Total charge per unit area induced in silicon

$$V_g - V_{fb} = V_{ox} + \phi_s = \frac{-Q_s}{C_{ox}} + \phi_s$$

Taur, Eq. (4.7)

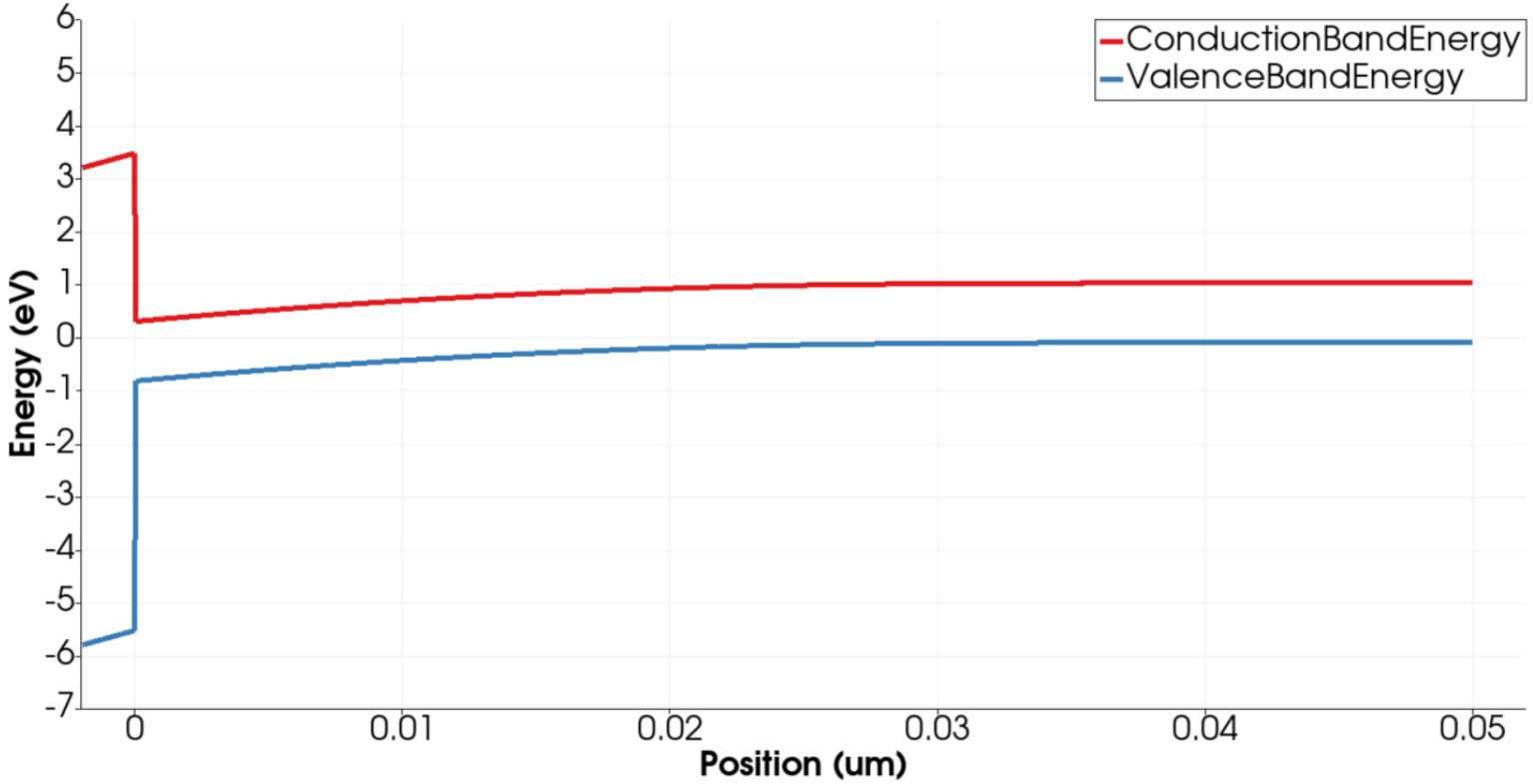
  - C_{ox} is the oxide capacitance per unit area,

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

Taur, Eq. (4.3)

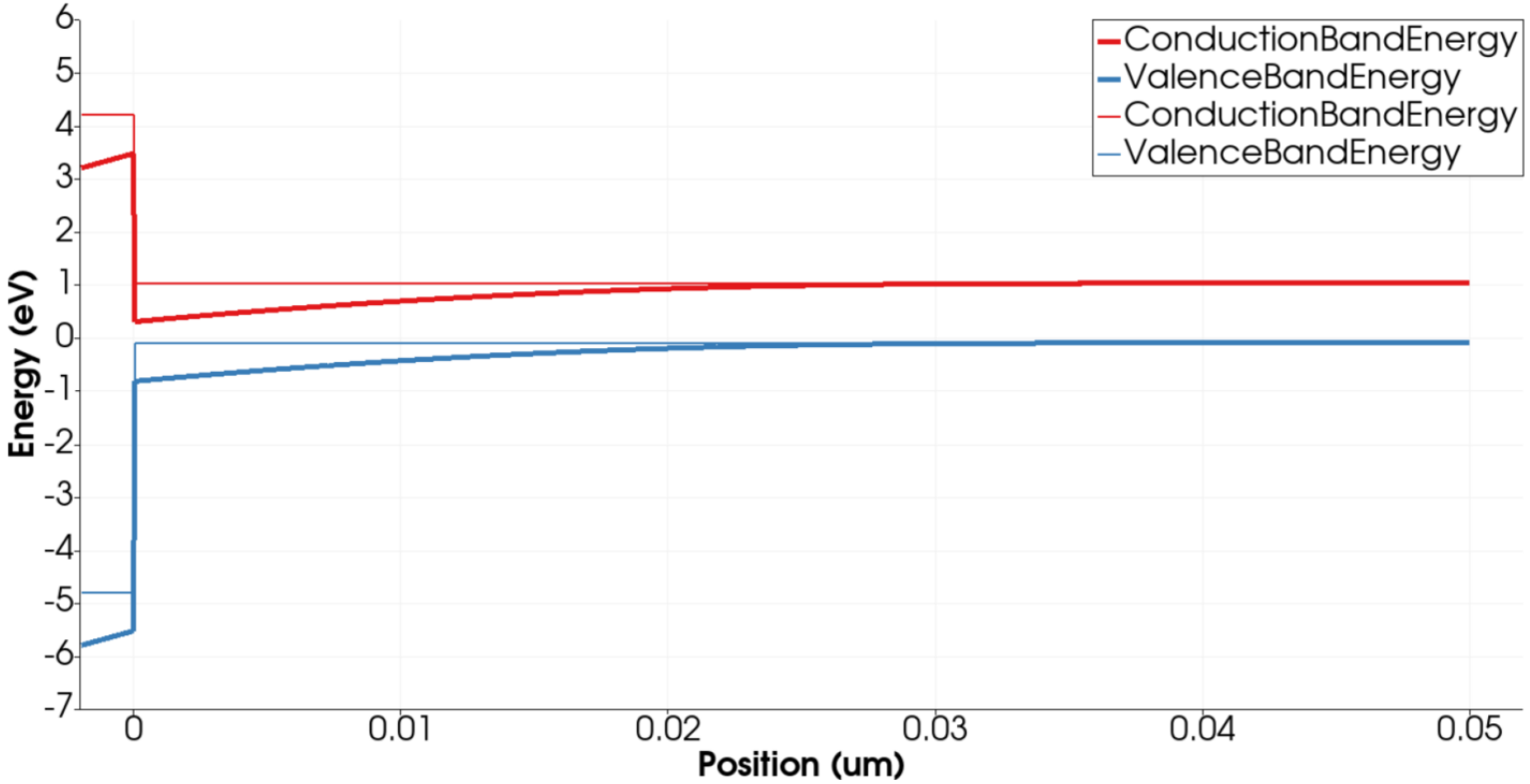
A model MOS @ 0 V

- Gate workfunction = 4.1 eV. 2-nm-thick oxide. 10^{18} cm^{-3}.
  - Depletion



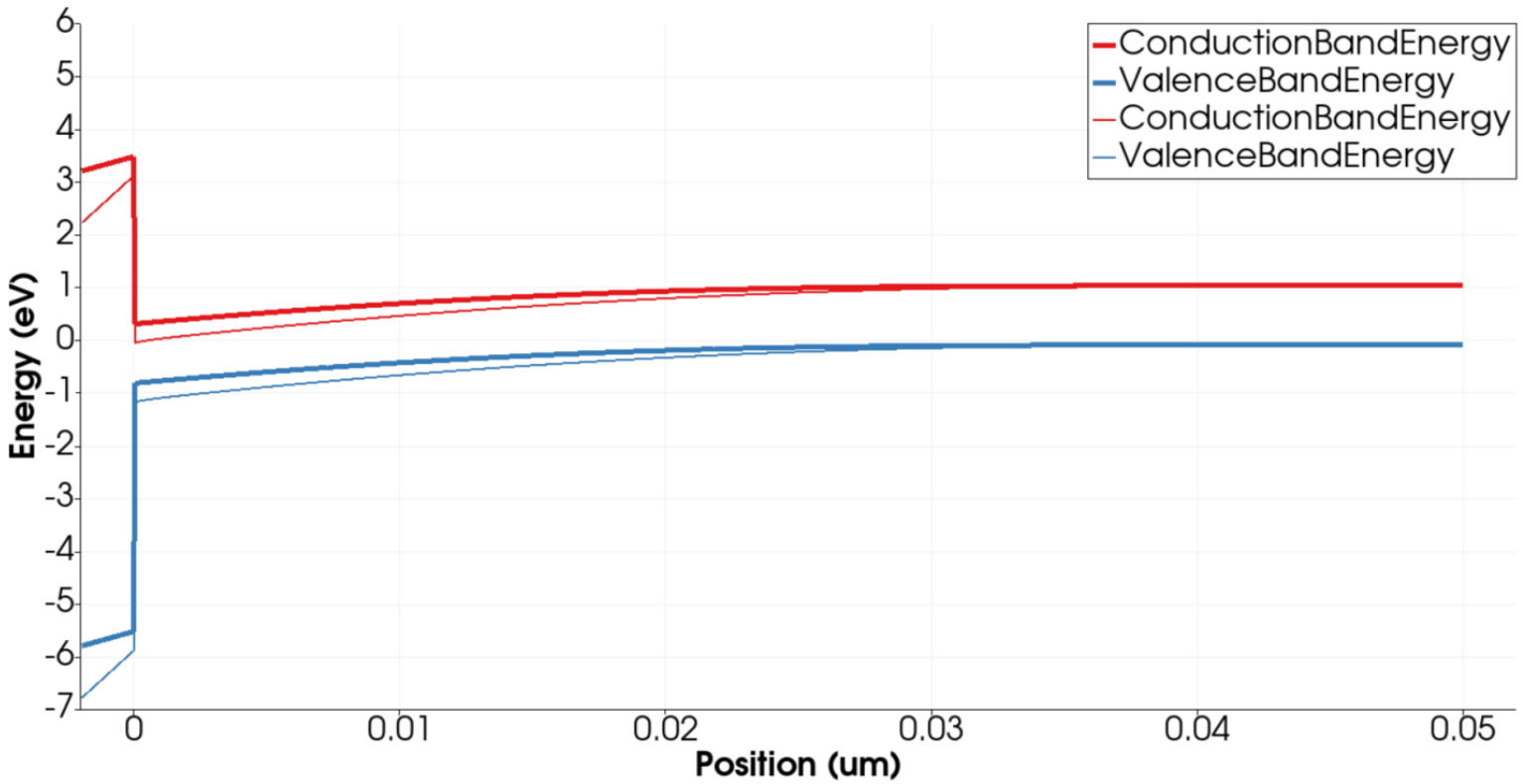
A model MOS @ flatband voltage

- The flatband voltage is -1.01 V.
  - In this case, the band energies are flat.



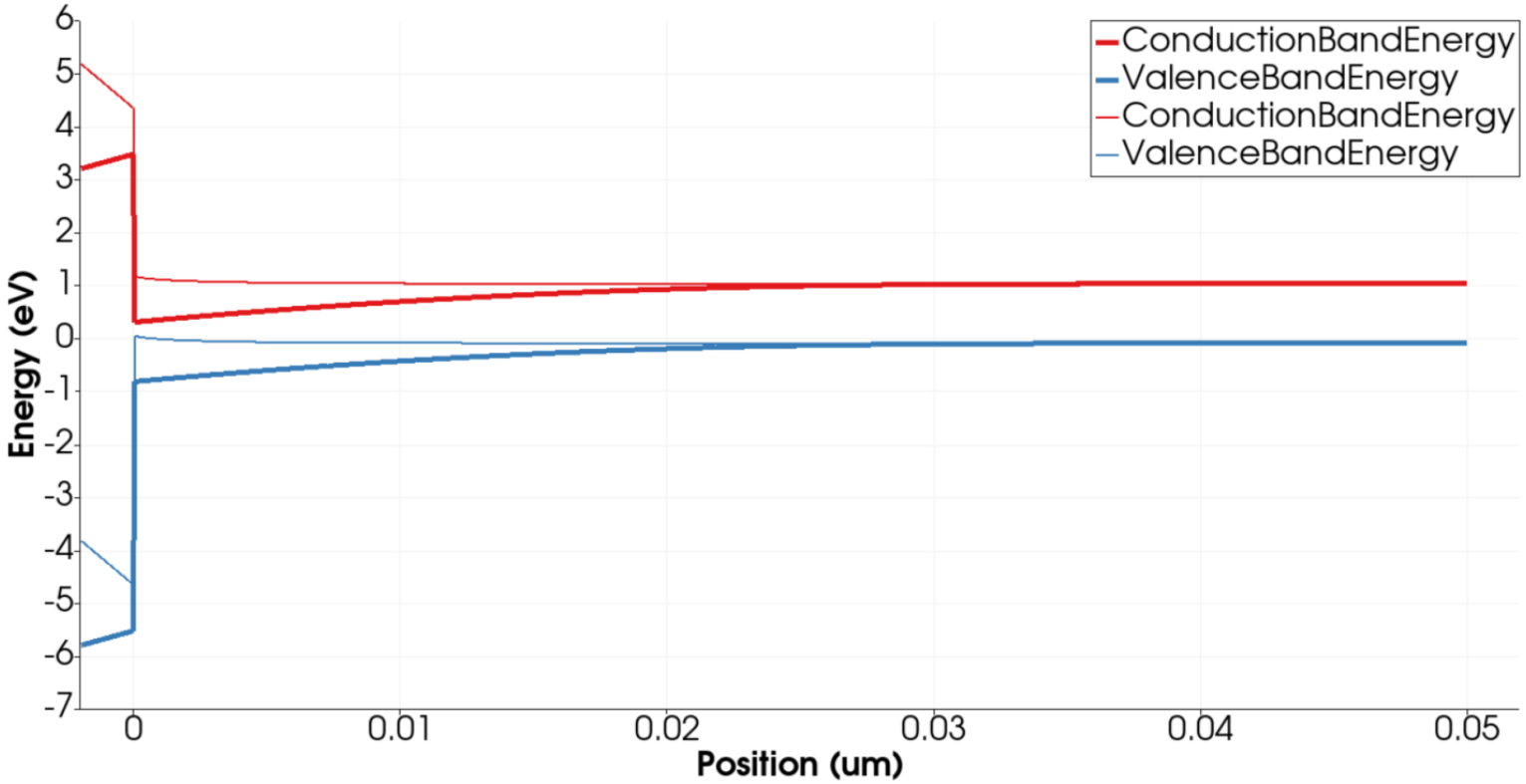
A model MOS @ 1 V

- Further downward band bending
  - Inversion. Increase of surface potential



A model MOS @ -2 V

- Accumulation
  - Upward band bending. A lot of holes



General Q_s-ϕ_s relation (1)

Inversion charge

- A generation relation for $Q_s = Q_d + Q_i$
  - The Poisson equation

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_{si}}[p(x) - n(x) - N_a]$$

Taur, Eq. (4.9)

General Q_s-ϕ_s relation (2)

- Following Prof. Taur's notation,

For a while, $\phi(\infty) = -\phi_B$ is used as the reference value. Therefore,

$$n(x) = n_i \exp\left(\frac{q\phi(x)}{k_B T}\right) \rightarrow n(x) = n(\infty) \exp\left(\frac{q\phi(x)}{k_B T}\right)$$ Taur, Eq. (4.12)

$$p(x) = n_i \exp\left(-\frac{q\phi(x)}{k_B T}\right) \rightarrow p(x) = p(\infty) \exp\left(-\frac{q\phi(x)}{k_B T}\right)$$ Taur, Eq. (4.11)

– The Poisson equation

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_{si}}\left[N_a\left(\exp\left(-\frac{q\phi}{k_B T}\right) - 1\right) - \frac{n_i^2}{N_a}\left(\exp\left(\frac{q\phi}{k_B T}\right) - 1\right)\right]$$

Taur, Eq. (4.13)

General Q_s-ϕ_s relation (3)

- Multiplying $\frac{d\phi}{dx}dx$,
  - The Poisson equation

$$\frac{d\phi}{dx}d\left(\frac{d\phi}{dx}\right) = -\frac{q}{\epsilon_{si}}\left[N_a\left(\exp\left(-\frac{q\phi}{k_BT}\right)-1\right)-\frac{n_i^2}{N_a}\left(\exp\left(\frac{q\phi}{k_BT}\right)-1\right)\right]d\phi$$

  - Integrate the above equation.

$$\int_0^{-E_x(x)}\frac{d\phi}{dx}d\left(\frac{d\phi}{dx}\right) = -\frac{q}{\epsilon_{si}}\int_0^{\phi(x)}\left[N_a\left(\exp\left(-\frac{q\phi}{k_BT}\right)-1\right)-\frac{n_i^2}{N_a}\left(\exp\left(\frac{q\phi}{k_BT}\right)-1\right)\right]d\phi$$

Taur, Eq. (4.14)

General Q_s-ϕ_s relation (4)

- (Square of) Electric field
  - From $\frac{1}{2}E_x^2(x) = -\frac{q}{\epsilon_{si}}\left[-N_a\frac{k_BT}{q}\exp\left(-\frac{q\phi}{k_BT}\right) - N_a\phi + N_a\frac{k_BT}{q} - \frac{n_i^2}{N_a}\frac{k_BT}{q}\exp\left(\frac{q\phi}{k_BT}\right) + \frac{n_i^2}{N_a}\phi + \frac{n_i^2}{N_a}\frac{k_BT}{q}\right]$, we get

$$E_x^2(x) = \frac{2k_BTN_a}{\epsilon_{si}}\left[\left(\exp\left(-\frac{q\phi}{k_BT}\right) + \frac{q\phi}{k_BT} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi}{k_BT}\right) - \frac{q\phi}{k_BT} - 1\right)\right]$$

Taur, Eq. (4.15)

General Q_s-ϕ_s relation (5)

- At $x = 0$, we have $\phi(0) = \phi_s$.
  - Then,

$$E_s^2 = \frac{2k_B T N_a}{\epsilon_{si}}\left[\left(\exp\left(-\frac{q\phi_s}{k_B T}\right) + \frac{q\phi_s}{k_B T} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_B T}\right) - \frac{q\phi_s}{k_B T} - 1\right)\right]$$

General Q_s-ϕ_s relation (6)

- At $x = 0$, we have $\phi(0) = \phi_s$.
  - From $Q_s = -\epsilon_{si} E_s$,

$$Q_s = \pm\sqrt{2\epsilon_{si} k_B T N_a}\left[\left(\exp\left(-\frac{q\phi_s}{k_B T}\right) + \frac{q\phi_s}{k_B T} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_B T}\right) - \frac{q\phi_s}{k_B T} - 1\right)\right]^{1/2}$$

Taur, Eq. (4.16)

Homework#4

- Due: 08:00 on March 18
- Submit a PDF report through the GIST LMS system.
  - 1) First, draw calculate Q_s (or E_s) as a function of ϕ_s.
  - Use the parameters for our example. (Gate workfunction = 4.1 eV. 2-nm-thick oxide. 10^{18} cm^{-3}.)

  - 2) From E_s, you can also calculate $V_{ox} = t_{ox}E_{ox} = t_{ox}\frac{\epsilon_{si}}{\epsilon_{ox}}E_s$.
  - Remember that

$$V_g - V_{fb} = \phi_s + V_{ox}$$

  - Draw Q_s as a function of V_g.

Thank you!